Positivity Properties of Fourier Expansions of Local Heights on Abelian Varieties

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Elliptic Curves

Local Heights on Elliptic Curves

$(K, v) =$ a complete field,

e.g., $K = \mathbb{C}$ or $K = \mathbb{C}_p$ or $K/\mathbb{Q}_p$

$E/K =$ an elliptic curve, say $y^2 = x^3 + ax + b$

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A (normalized) **canonical local height on** E/K is a function

$$\hat{\lambda} : E(\overline{K}) \smallsetminus \{O\} \longrightarrow \mathbb{R}$$

satisfying a functional equation and having a logarithmic pole at O:

- $\hat{\lambda}(2P) = 4\hat{\lambda}(P) + v\big(2y(P)\big) - \frac{1}{4}v\big(\Delta(E)\big)$
- $\hat{\lambda}(P) = \frac{1}{2}v^+\big(x(P)^{-1}\big) + O(1)$

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If K is non-archimedean and E/K has good reduction, then $\hat{\lambda}$ is given by the v-adic intersection $O \cdot P$ on the Néron model; or more explicitly, by the formula

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Note that $(*)$ is non-negative, an often useful fact.

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When E/K has bad reduction, there is a correction term that may be negative. The hero of this talk is the correction term.

Local Heights for Split Mulitplicative Reduction

Going to an extension field if necessary, we will assume:

E/K has split multiplicative (aka split semi-stable) reduction.

Elliptic Curves with Totally Multiplicative Reduction

We fix:

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$$u : E(\overline{K}) \longrightarrow \overline{K}^*/q^{\mathbb{Z}}$$

to define a homomorphism

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Alternative Description: Let $N = \operatorname{ord}_v(q)$. Then

$$t : E(\overline{K}) \longrightarrow E(\overline{K})/E_0(\overline{K}) \cong \tfrac{1}{N}\mathbb{Z}/\mathbb{Z},$$

where $E(\overline{K})/E_0(\overline{K})$ is the group of components on the special fiber of the Néron model of E.

Canonical Local Height on an Elliptic Curve

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It is given explicitly by the formula

$$\hat{\lambda}(P) = \frac{1}{2}\mathbb{B}_2(t_P) + \underbrace{\text{“intersection stuff”}}_{\text{This quantity is non-negative.}},$$

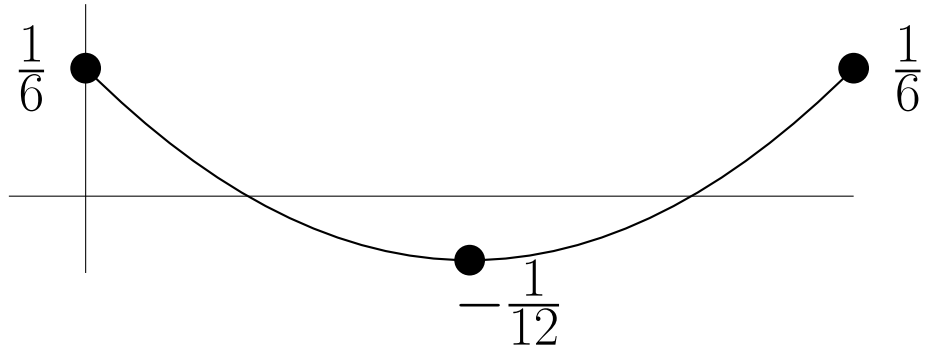
where

$$\mathbb{B}_2(T) = \underbrace{T^2 - T + \frac{1}{6}}_{\text{The periodic 2nd Bernoulli polynomial.}} \quad \text{on} \quad [0,1].$$

Some Remarks on Elliptic Curve Local Heights

- The periodic 2nd Bernoulli polynomial takes both positive and negative values. Its average value is 0.

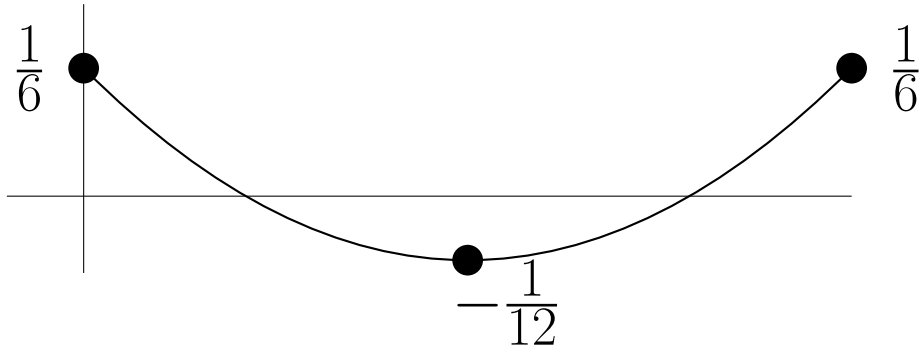
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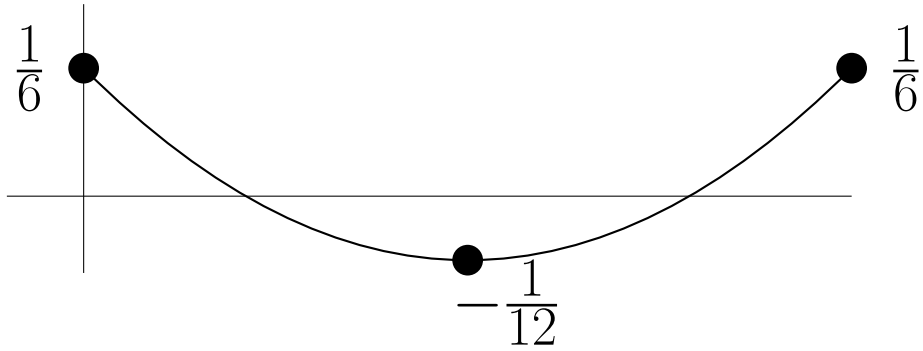
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$$\mathbb{B}_2(t) = \operatorname{dist}\left(t,\, \tfrac{1}{2}+\mathbb{Z}\right)^2 - \tfrac{1}{12}.$$

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- There is a similar formula for $\hat{\lambda}$ when $K = \mathbb{C}$ involving a Greens function. In general, if $\mathcal{K}/\mathbb{Q}$ is a number field and $\mathcal{E}/\mathcal{K}$ an elliptic curve, then

$$\hat{h}_{\mathcal{E}}(\mathcal{P}) = \sum_{v\in M_{\mathcal{K}}} n_v \hat{\lambda}_v(\mathcal{P}) \quad \text{for all } \mathcal{P}\in\mathcal{E}(\mathcal{K})\smallsetminus\{\mathcal{O}\}.$$

Non-negative Weighted Bernoulli Averages?

To prove lower bounds for the canonical height, it would be very useful to have weights $w_1, w_2, \ldots, w_M$ so that

$$\sum_{m=1}^{M} w_m \cdot \mathbb{B}_2(mt) \geq 0 \quad \text{for all } t \in \mathbb{R}.$$

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The key to proving a result of this sort lies in the Fourier expansion of $\mathbb{B}_2(t)$.

Fourier Averaging for $\mathbb{B}_2$

The Fourier expansion of the periodic function

$$\mathbb{B}_2 : \mathbb{R}/\mathbb{Z} \longrightarrow \mathbb{R}, \qquad \mathbb{B}_2(T) = T^2 - T + \frac{1}{6},$$

is given by

$$\mathbb{B}_2(t) = \sum_{0 \neq n \in \mathbb{Z}} \frac{1}{2\pi^2 n^2} \cdot e^{2\pi i n t}.$$

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Key Fact: Positive Fourier coeffiients!

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Fourier Averaging for $\mathbb{B}_2$

We use the Fourier expansion in a Fejér kernel weighted sum:

$$\sum_{m=1}^{M} \underbrace{\left(1 - \frac{m}{M+1}\right)}_{\text{Fejér kernel}} \mathbb{B}_2(mt)$$

$$= \sum_{0 \neq n \in \mathbb{Z}} \frac{1}{2\pi^2 n^2} \left(\frac{1}{M+1} \left| \sum_{m=0}^{M} e^{2\pi i m n t} \right|^2 - 1 \right)$$

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Then the positivity of the Fourier coefficients gives the non-trivial lower bound

$$\underbrace{\sum_{m=1}^{M} \left(1 - \frac{m}{M+1}\right) \mathbb{B}_2(mt)}_{\text{A sum of } M \text{ potentially negative terms.}} \geq -\frac{1}{12}.$$

Generalizing to Abelian Varieties

Notation

$K =$ a global field

$A/K =$ an abelian variety of dimension g

$\Theta \in \mathrm{Div}(A)$, an ample effective symmetric divisor

$\phi_m \in K(A)$, a function satisfying

$$[m]^*\Theta = m^2\Theta + (\phi_m)$$

$\hat{h} : A(\overline{K}) \to \mathbb{R}_{\geq 0}$, canonical height associated to Θ

Local and Global Heights on Abelian Varieties

Theorem. For each $v \in M_{\overline{K}}$, there is a unique *local height*

$$\hat{\lambda}_v : A(\overline{K}_v) \smallsetminus |\Theta| \longrightarrow \mathbb{R}$$

satisfying:

- $\hat{\lambda}_v(mP) = m^2\hat{\lambda}_v(P) + v\big(\phi_m(P)\big) + c_v$
- $\hat{\lambda}_v(P) = \log|\operatorname{Dist}_v(P,\Theta)^{-1}|_v + O(1)$

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- $\hat{h}_\Theta(P) = \displaystyle\sum_{v\in M_{K(P)}} \hat{\lambda}_v(P) - \underbrace{\kappa(A/K)}_{\kappa(A/K) \text{ vanishes if } \dim(A)=1.}$

Local Height for Split Multiplicative Reduction

Suppose A has totally split multiplicative reduction at v. Then

$$\hat{\lambda}_v = \hat{\lambda}_v^{\mathbb{B}} + \hat{\lambda}_v^{\mathbb{I}}$$

splits into two pieces:

$$\hat{\lambda}_v^{\mathbb{B}} = \text{Bernoulli part of } \hat{\lambda}_v,$$
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We can describe $\hat{\lambda}_v^{\mathbb{B}}$ using the Mumford–Tate uniformization.

Local Height for Split Multiplicative Reduction

The Mumford–Tate uniformization gives a homomorphism

$$\boldsymbol{t}_v : A(K_v) \cong \mathbb{G}_m(K_v)^g/\Lambda_v \xrightarrow{\mathrm{ord}_v} (\mathbb{R}/\mathbb{Z})^g,$$

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where

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Goal: An explicit Fourier expansion for $\hat{\lambda}_v^{\mathbb{B}}(P)$.

S

Formulas for Abelian Surfaces

The Fourier Series of $L_Q = \inf Q$

Theorem: (Looper–JS)

Let $$Q(x,y) = ax^2 + 2bxy + cy^2 \in \mathbb{Z}[x,y]$$ be a positive definite quadratic form, which we normalize by $c \geq a > 2b \geq 0$. We consider the function

$$L_Q : (\mathbb{R}/\mathbb{Z})^2 \longrightarrow \mathbb{R},$$

$$L_Q(x,y) = \inf \left\{ Q(\xi,\eta) : \begin{matrix} \xi \in x + \frac{1}{2} + \mathbb{Z} \\ \eta \in y + \frac{1}{2} + \mathbb{Z} \end{matrix} \right\}.$$

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Write the Fourier expansion of L_Q as

$$L_Q(x,y) = \sum_{m,n \in \mathbb{Z}} \hat{L}_Q(m,n) e^{2\pi i(mx+ny)}.$$

The Fourier Series of $L_Q = \inf Q$

Theorem: (Looper–JS) (continued)
For most (m, n) the Fourier coefficient $\hat{L}_Q(m, n)$ equals

$$\frac{(ac - b^2)\sin\left(2\pi\dfrac{c(a-b)m + a(c-b)n}{2(ac-b^2)}\right)}{4\pi^3(cm - bn)(an - bm)\big((c-b)m + (a-b)n\big)}.$$

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$$cm = bn \neq 0 \quad\Longrightarrow\quad \hat{L}(m,n) = \frac{(-1)^n(a-b)c^2}{2\pi^2(ac-b^2)n^2},$$

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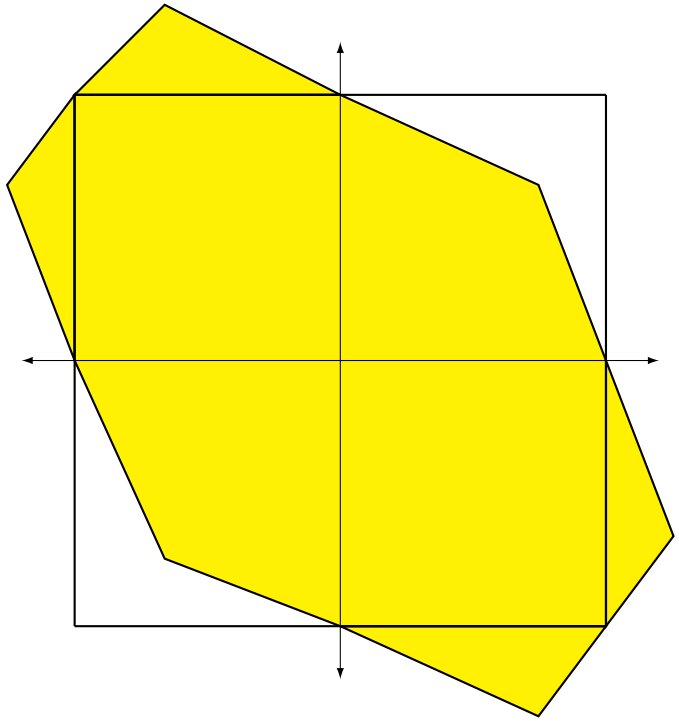
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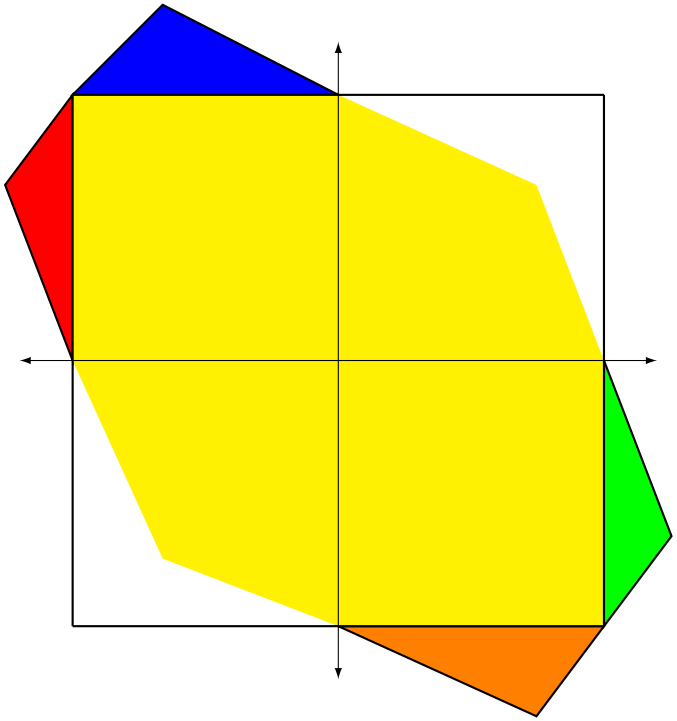
Further Note. If m and n are even, the special case coefficients are positive.

Computing the Fourier Coefficients $\hat{L}_Q$



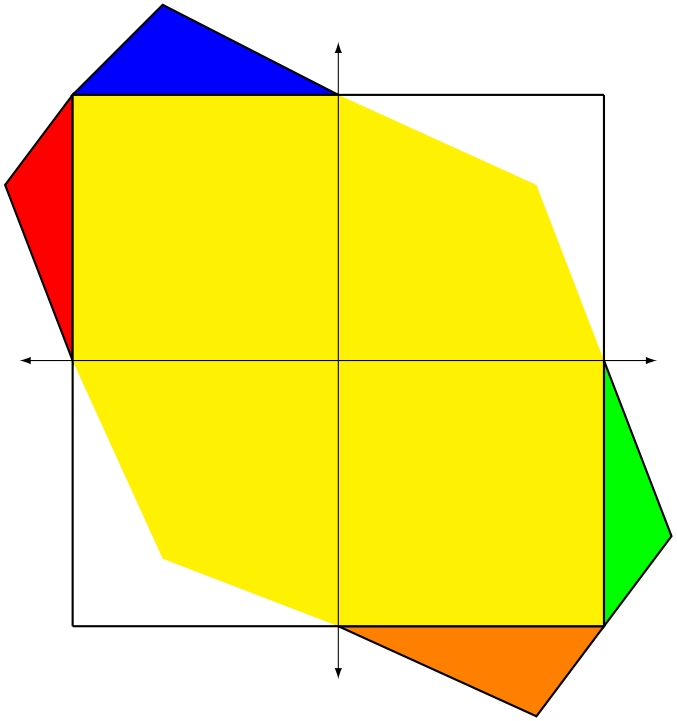
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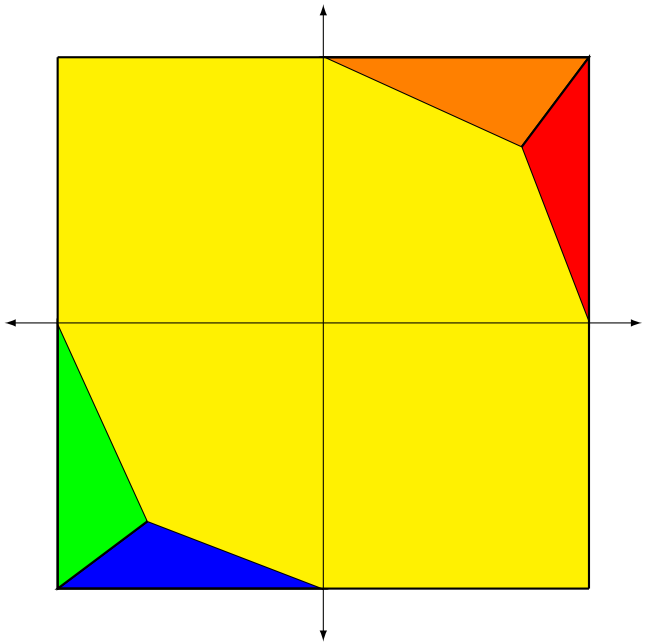


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The Fundamental Domain for $L_Q = \inf Q$

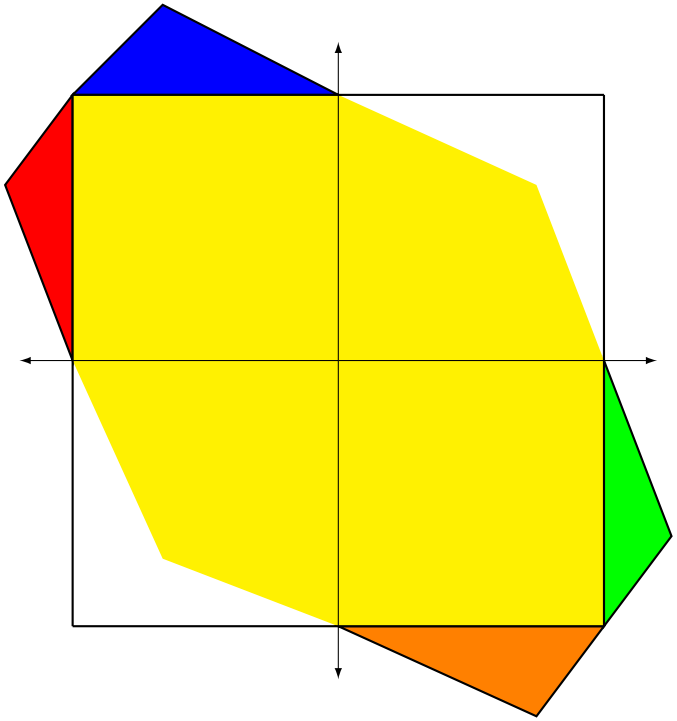


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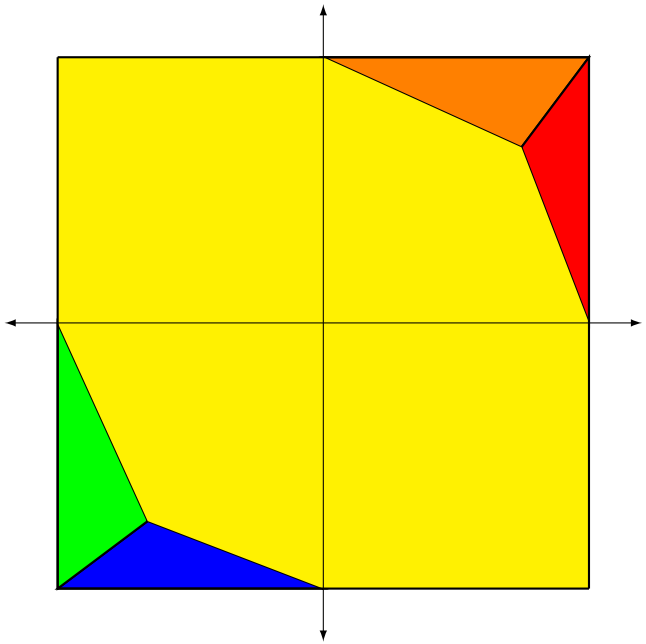


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The Fundamental Domain for $L_Q = \inf Q$



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Computing $\hat{L}_Q(m,n)$ is a painful multi-region integration!

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- **Application**: Nicole and I used this to study Lehmer's conjecture for abelian surfaces over function fields, although the normalization constant $\kappa > 0$ in the formula relating the sum of the local heights to the global height prevented a full resolution.
- **Question**: It seems to be an interesting problem to compute the Fourier coefficients for L_Q for lattices in $\mathbb{R}^3$, and more generally in $\mathbb{R}^g$.

Some Canonical Height Lower Bounds

Local decompositions and Fourier averaging is one approach to a fundamental height conjecture, which I will state both intuitively and precisely (but without defining all of the terms).

Conjecture (Informal). (Lang–JS)

(Canonical Height of Large-Orbit Point)
$$\gg \text{(Height of the Abelian Variety)}.$$

"If A is complicated, then its points are complicated."

Some Canonical Height Lower Bounds

Conjecture. (Lang–JS) For $\overline{\mathbb{Z}P} = A$,

$$\hat{h}_{A,\mathcal{L}}(P) \geq C\big(\dim A, [K:\mathbb{Q}]\big) \cdot h_{\text{Faltings}}(A/K).$$

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Theorems.

Number Fields:

(a) *abc* conjecture $\implies$ $\dim A = 1$ (Hindry–JS - 1988)

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These results are all about a single point.

Some Canonical Height Lower Bounds

How about if $A(K)$ has rank ≥ 2? Are there better lower bounds for the second smallest point? JitWu suggested an intriguing general conjecture, which in even the following simplified case seems very challenging.

Some Canonical Height Lower Bounds

Fix $g \geq 1$ and consider the set

$$\mathcal{A}_r(g) = \left\{ \begin{array}{c} \text{p.p. abelian varieties } A/\mathbb{Q} \text{ of} \\ \dim(A) = g \text{ with rank } A(\mathbb{Q}) \geq r \end{array} \right\}.$$

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We look at

$$\lambda_r(g) = \liminf_{\substack{A \in \mathcal{A}(g) \\ P_1,\ldots,P_r \in A(\mathbb{Q}) \\ \text{independent} \\ \overline{\mathbb{Z}P_i} = A}} \left\{ \frac{\max\{\hat{h}_A(P_1), \ldots, \hat{h}_A(P_r)\}}{h_{\text{Faltings}}(A/\mathbb{Q})} \right\}.$$

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Even $\lambda_2(1) > \lambda_1(1)$ seems to require new ideas.

Concluding with an Amusing ChatGPT Conversation

$B_2(x) = (x - \frac{1}{2})^2 - \frac{1}{12}$ has positive Fourier coefficients.
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ChatGPT: You're right — it is **not** correct. Thank you for catching that. I made an error when solving for the coefficients by hand. ... It's very easy to make arithmetic mistakes doing this manually.

In Conclusion









I want to thank you for your attention and the organizers for the invitation to speak.









Positivity Properties of Fourier Expansions of Local Heights on Abelian Varieties

Joseph H. Silverman
Joint work with Nicole Looper

Brown University

Workshop on Arakelov and Diophantine Geometry
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